

Ten Layers of Signed Completion and Upper Counterloops: A Typed Separation Review from Actual Direct Tails to Deterministic Transform Phases

Bin Wang

August 2026

Abstract

We audit RH-352–RH-360 and prove a typed separation theorem for the batch. RH-352–RH-354 are actual results for the direct coefficient $p = \tau - a$ and actual signed completion remainders, but their closure statements remain selected and normalized. RH-355–RH-360 are unconditional results for the deterministic graded counterloop and its normalized budgets; every actual-head inheritance statement in those papers assumes the same unproved, unnormalized, same-clock hypothesis

$$D_{4k}(R) = \sum_{2 \leq n < 4k} \frac{|h_{\sigma,n} - s_{k,n}| R^n}{n} \longrightarrow 0.$$

The exact source identities are

$$p_n = q_n - d_n, \quad d_n = h_n - s_n.$$

We isolate their coefficient fiber: for fixed p and s , every signed array e gives $d = e$, $q = p + e$, and $h = s + e$. Thus p and s alone do not determine q , h , or their weighted budgets. This is an algebraic information-class theorem, not a construction of noisy operators. It forbids promotion of RH-354’s normalized p tail to the full-trace q/E_{off} ledger without defect transport, and forbids interpreting the exponentially large deterministic s budgets as actual spectral submultisets. The deterministic terminal-lag route is closed at its declared budget type through the transform phase diagram: RH-358 gives geometric localization, RH-359 gives logarithmic inverse accuracy, and RH-360 gives the three-regime exponential-transform phase diagram. The nine upstream result files contain 45/45 false Gate values and 129/129 false forbidden-claim values. RH-241 and RH-288 remain open/inactive, Gates A–E remain false, and no Hilbert–Polya, zero, divisor, or RH conclusion follows.

1 The two exact coefficient types

The batch uses one bounded-phase physical clock and the Hardy normalization fixed upstream. The actual modulus-complement trace and deterministic numerator anchor define

$$p_{\sigma,k,n} = \tau_{\sigma,n} - a_n. \tag{1}$$

RH-340 identifies the same coefficient with the difference between a Hardy full-trace coefficient and the head/counterloop defect

$$\boxed{p_{\sigma,k,n} = q_{\sigma,k,n} - d_{\sigma,k,n}}, \quad \boxed{d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}}. \tag{2}$$

Here $h_{\sigma,n}$ is the moment of the actual modulus-complete Hardy head, while $s_{k,n}$ is the moment of the deterministic graded counterloop [9, 13]. Thus, suppressing the common clock,

$$q = p + d, \quad h = s + d. \tag{3}$$

Equations (2)–(3) are identities of coefficient arrays. They do not assert a spectral inclusion between the counterloop and the actual head.

The deterministic counterloop itself is the finite multiset

$$\mathcal{Y}_k = \{\beta_k e^{ij\pi/k}, \beta_k e^{-ij\pi/k} : 1 \leq j \leq k-1\}, \quad (4)$$

whose exact moments are

$$s_{k,n} = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n). \quad (5)$$

It is a source-locked auxiliary graded shell. The actual head $\mathcal{H}_\sigma$ is a different object. Even the conditional moment transport derived later in the batch is not a theorem that (4) is a submultiset of $\mathcal{H}_\sigma$.

2 Ten source-locked layers

The following table retains the object type of every result. “Actual” means an identified coefficient or signed remainder of the noisy physical construction. “Deterministic” means the finite graded counterloop or a normalized budget built solely from its exact moments.

paper	strict result	retained boundary
RH-352	Actual normalized direct cancellation on every lower-even window $J_k \rightarrow \infty$, $J_k = o(k)$, and uniform actual Y tracking of $S - \mathcal{P}$.	The unnormalized separate noisy majorant is superunit; critical, first-lower, odd, upper, and full-prefix channels are not closed.
RH-353	Actual critical/first-lower normalized completion laws and the phase-free gap $Z_k^0 - \lambda Z_k^- \rightarrow 2 - \lambda$.	The selected Y supply is not a lower bound for the direct coefficients p_k^0, p_k^- and is not a full-prefix theorem.
RH-354	Actual parity-free normalized direct tail for $p = \tau - a$ above $N_k = 2k - L_k$, including both parities and the strict upper band.	The result is not a theorem for q, d , or E_{off} ; low orders and the unnormalized prefix remain open.
RH-355	Exact deterministic strict-upper burden, terminal share, and the precision that $D_{4k}(R) \rightarrow 0$ would force.	Actual-head conclusions are conditional; a normalized-shell counterexample is information-class only.
RH-356	Deterministic mesoscopic post-alias crossover, fixed-depth factor, and complete logarithmic floor phase.	Actual inheritance assumes $D_{4k}(R) \rightarrow 0$; no linear-depth theorem is claimed here.
RH-357	Deterministic uniform endpoint profile over $1 \leq L \leq k-1$, including C_M drift, denominator correction, and rational/irrational phases.	Actual budget inheritance remains conditional and gives no root or rank identification.
RH-358	Deterministic terminal-lag tail profile, geometric total-variation localization, first two moments, and the exact diverging-window criterion.	Actual weighted lags inherit only under $D_{4k}(R) \rightarrow 0$; the lag integer is not the full-trace coefficient q .
RH-359	Deterministic logarithmic accuracy law, full cluster interval, and minimal integer-window correction set $[0, 1]$.	Actual accuracy inherits only under the same open transport leaf.
RH-360	Deterministic generating-function transition below, at, and above $z = x$, together with three tilted budget laws.	The probabilities are not eigenvalue or root-counting measures; actual transfer is conditional.
RH-361	Exact typed batch separation and coefficient-fiber nonpromotion theorem.	Information-class logic only; no physical counterexample, root, rank, determinant, or Gate theorem is constructed.

The source results are [1, 2, 4–7, 10, 11, 13].

3 The actual selected and normalized branch

Fix

$$H_m = mR^{-2m}, \quad x = (\beta R)^2 > 1, \quad m_{k,j} = k - j, \quad 2 \leq j \leq J_k, \quad (6)$$

where $J_k \rightarrow \infty$ and $J_k = o(k)$. RH-352 uses the exact actual coefficient identity

$$p_{k,j} = Y_{k,j} + \mathcal{P}_{k,j} - S_{k,j} \quad (7)$$

and the all-order source caps for τ and a to prove

$$\limsup_{k \rightarrow \infty} \left(\sup_{2 \leq j \leq J_k} \frac{|p_{k,j}|}{2H_{m_{k,j}} x^{m_{k,j}}} \right)^{1/k} \leq \max \left\{ \frac{r_H^2 \lambda^3}{4}, \frac{1}{\lambda} \right\} < 1. \quad (8)$$

Solving (7) then makes the actual Y array track $S - \mathcal{P}$ uniformly at the same local natural scale. This is a genuine moving-order actual theorem. Removing the factor x^m is not allowed: the separate noisy bound then has certified root

$$\lambda^2 (q_0 R)^2 > \frac{9604}{7225} > 1, \quad q_0 = \frac{1}{2}. \quad (9)$$

The inequality is a failure of that absolute-majorant method, not a lower bound for the signed difference p .

RH-353 treats the two boundary coordinates excluded from the RH-352 ladder. For the actual normalized signed remainders

$$Z_k^0 = \frac{C_M Y_k^0}{2H_k x^k}, \quad Z_k^- = \frac{C_M Y_k^-}{2H_{k-1} x^{k-1}}, \quad (10)$$

it proves

$$Z_k^0 - \lambda Z_k^- \longrightarrow 2 - \lambda, \quad \liminf \max\{|Z_k^0|, |Z_k^-|\} \geq \frac{2 - \lambda}{1 + \lambda} > \frac{1}{9}. \quad (11)$$

The resulting selected unnormalized Y supply cancels the leading packets; it does not imply divergence of either direct coefficient.

Finally RH-354 removes parity and order restrictions above a moving cut. If $N_k = 2k - L_k$ and $L_k = o(k)$, its actual bottom-normalized direct tail

$$\mathfrak{W}_k = x^{-N_k/2} \sum_{n \geq N_k} |p_{\sigma,k,n}| R^n \quad (12)$$

has a subunit root ceiling. The corresponding alias-normalized logarithmic tail

$$x^{-k} \sum_{n \geq N_k} \frac{|p_{\sigma,k,n}| R^n}{n} \quad (13)$$

also decays. Equations (12)–(13) are the broadest actual direct theorem in the batch, but their object is still p , not the full-trace coefficient q in (2).

4 The deterministic upper-counterloop branch

Put

$$y_k = (\beta_k R)^2 = x \exp \left[-\frac{\log C_M}{k} + o(k^{-1}) \right]. \quad (14)$$

For $1 \leq L \leq k - 1$, the exact first-alias and post-alias budgets are

$$A_k = (1 - k^{-1}) y_k^k, \quad B_k(L) = \sum_{j=1}^L \frac{y_k^{k+j}}{k+j}. \quad (15)$$

RH-355–RH-357 determine (15) on every depth scale. In particular,

$$B_k(L) = \frac{x^{k+L+1}(1-x^{-L})}{C_M^{1+L/k}(k+L)(x-1)}(1+o(1)) \quad (16)$$

uniformly over the complete strict upper band. At $L = k - 1$, this gives the RH-355 burden

$$B_k(k-1) \sim \frac{x^{2k}}{2C_M^2 k(x-1)}. \quad (17)$$

These are exact asymptotics for the deterministic moments (5). Their exponential size does not identify an actual head moment or actual spectral atom.

RH-358 reverses the upper band. With $C_k = B_k(k-1)$ and terminal-lag probability

$$\pi_k(r) = \frac{y_k^{2k-1-r}/(2k-1-r)}{C_k}, \quad 0 \leq r \leq k-2, \quad (18)$$

it proves

$$\left\| \pi_k - (1-x^{-1})x^{-r} \right\|_{\ell^1} \longrightarrow 0. \quad (19)$$

RH-359 inverts the terminal tail $E_k(q)$: for $q_k = \lfloor a \log_x k + c \rfloor$,

$$k^a E_k(q_k) = x^{\{a \log_x k + c\} - c} (1 + o(1)), \quad (20)$$

and the minimal admissible window differs from $a \log_x k + c$ by a sequence with complete limit set $[0, 1]$.

RH-360 completes the deterministic route with

$$G_k(z) = \sum_{r=0}^{k-2} z^r \pi_k(r). \quad (21)$$

For fixed $z < x$, the transform tends to $(1-x^{-1})/(1-z/x)$. In the critical window $z_k = x \exp(\tau/k)$,

$$\frac{G_k(z_k)}{k} \longrightarrow 2(1-x^{-1}) \int_0^1 \frac{e^{(\tau+\log C_M)s}}{2-s} ds, \quad (22)$$

while for fixed $z > x$,

$$G_k(z) \sim \frac{2C_M(1-x^{-1})}{1-x/z} \left(\frac{z}{x}\right)^{k-2}. \quad (23)$$

Thus the deterministic terminal-lag sequence closes through its entire transform phase diagram. The tilted laws remain normalized budget laws, not eigenvalue distributions or noisy stochastic laws.

5 The typed coefficient-fiber theorem

The exact reason that the two branches do not glue is algebraic before it is analytic.

Definition 5.1 (finite typed coefficient information class). Let I be a nonempty finite order set and let $w_n > 0$. Fix two signed arrays $(p_n)_{n \in I}$ and $(s_n)_{n \in I}$. A typed coefficient fiber is an array $(e_n)_{n \in I}$ together with

$$d_n^{[e]} = e_n, \quad q_n^{[e]} = p_n + e_n, \quad h_n^{[e]} = s_n + e_n. \quad (24)$$

No operator, determinant, rank, root set, or cross-order realizability condition is included in this information class.

For an array u , write

$$\|u\|_{1,w} = \sum_{n \in I} w_n |u_n|. \quad (25)$$

Theorem 5.2 (exact typed fiber and nonpromotion). *Every array e in (24) satisfies*

$$d^{[e]} = h^{[e]} - s, \quad p = q^{[e]} - d^{[e]} \quad (26)$$

coordinatewise. If v is any nonzero signed array, then along $e^{(t)} = tv$,

$$\|q^{[e^{(t)}]}\|_{1,w} \longrightarrow \infty, \quad \|h^{[e^{(t)}]}\|_{1,w} \longrightarrow \infty \quad (|t| \rightarrow \infty), \quad (27)$$

while p and s remain fixed. Moreover, the choices $e = -p$ and $e = -s$ give $q^{[-p]} = 0$ and $h^{[-s]} = 0$, respectively. Consequently, as e ranges over this coefficient information class, neither $\|q^{[e]}\|_{1,w}$ nor $\|h^{[e]}\|_{1,w}$ has a finite uniform upper bound determined by p, s , while the infimum of each norm is zero.

Proof. Substitution of (24) gives $h^{[e]} - s = e = d^{[e]}$ and $q^{[e]} - d^{[e]} = (p + e) - e = p$. Since $v \neq 0$ and all weights are positive, $\|tv\|_{1,w} = |t|\|v\|_{1,w} \rightarrow \infty$. The reverse triangle inequality gives

$$\|p + tv\|_{1,w} \geq |t|\|v\|_{1,w} - \|p\|_{1,w}, \quad \|s + tv\|_{1,w} \geq |t|\|v\|_{1,w} - \|s\|_{1,w},$$

which proves (27). Finally, $q^{[-p]} = p - p = 0$ and $h^{[-s]} = s - s = 0$. $\square$

Remark 5.3 (logical scope). The theorem does not say that every fiber is realized by a noisy quadratic operator. It says that the coefficient identities and the fixed arrays p, s do not by themselves imply the missing conclusion. A physical promotion requires an additional theorem restricting the actual defect $d = h - s$.

Corollary 5.4 (two forbidden promotions). *The following deductions are unavailable from the batch premises alone.*

1. *A small selected or normalized actual direct budget for p cannot be promoted to the full-trace q/E_{off} budget without a same-clock bound for d .*
2. *A large deterministic counterloop budget for s cannot be promoted to an actual-head budget, root set, rank law, or spectral-submultiset statement.*

Proof. The two budget nonpromotions follow from theorem 5.2: the first uses $q = p + d$, and the second uses $h = s + d$. The further E_{off} , root, rank, and multiset deductions are not encoded by the fiber premises and are absent from the source-locked table. They require a separate realization or identification theorem that is unavailable in the batch. $\square$

6 The batch separation theorem

Theorem 6.1 (RH-352–RH-361 typed batch separation). *The ten layers satisfy all of the following.*

1. *RH-352–RH-354 give unconditional actual conclusions only for selected natural/alias normalizations of p and Y . RH-353 also converts its normalized gap into a selected signed- Y supply, not a direct or full-prefix lower bound.*
2. *RH-355–RH-360 give unconditional conclusions only for the deterministic counterloop s and normalized budgets derived from it.*
3. *Every actual-head transfer in RH-355–RH-360 assumes the same original unnormalized, same-clock leaf $D_{4k}(R) \rightarrow 0$. No paper in the batch proves that leaf.*
4. *The deterministic terminal-lag route is complete through (22)–(23), but no actual root, rank, spectrum, determinant, q , or E_{off} conclusion follows.*
5. *RH-241’s moving noisy all-order envelope and coefficient bridge remain open, and RH-288 remains inactive.*

Proof. Items 1–3 follow from the exact object and hypothesis columns in the source-locked table and from the defining statements of the nine upstream papers. The distinction is also enforced by (2): RH-354 controls p , whereas the full-trace ledger uses $q = p + d$. By theorem 5.2 and corollary 5.4, neither q nor the actual head is determined without a theorem for d .

Item 4 is the chain RH-358–RH-360 summarized in (19)–(23). The sources explicitly define these probabilities from normalized deterministic absolute budget weights, so no spectral interpretation is licensed. Item 5 is the retained frontier of RH-241 and the typed prefix requirement of RH-288 [3, 8, 12]. $\square$

7 Executable audit and claim boundary

The executable artifact reads the nine upstream result ledgers and checks the exact forbidden-claim vector

$$(15, 14, 13, 14, 14, 14, 14, 15, 16), \quad \sum = 129. \quad (28)$$

All 129 values are false. Each upstream file also contains five false Gate values, for 45/45 false values. RH-361 adds five false Gate values and twenty false forbidden claims, so the ten-paper totals are 50/50 and 149/149, respectively.

The finite rows instantiate theorem 5.2 with rational arrays and positive rational weights. They check both identities and exhibit different q/h budgets at fixed p/s . They are formula-reproduction checks only: they do not construct physical heads or noisy operators and are not asymptotic evidence.

The individual archive contains twenty RH-361 publication files. The nine upstream manifests contain 156, giving 176 files in the batch manifest. All manifests are SHA-256 rehashed by independent verification scripts.

The narrowest positive continuation is an actual theorem proving the unnormalized same-clock $D_{4k}(R) \rightarrow 0$ leaf, or a typed direct theorem for q/E_{off} that bypasses it. An unnormalized complete direct-prefix theorem and the low orders below the RH-354 cut remain separate. The RH-241 moving noisy all-order envelope and no-over-extraction bridge remain open. RH-288 may be activated only after the complete direct prefix and both analytic tails close in one physical determinant data type.

Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt-weighted prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

- [1] Bin Wang. A phase-free critical–first-lower actual signed-completion gap, 2026. Repository paper RH-353.
- [2] Bin Wang. A parity-free near-alias actual direct-tail envelope, 2026. Repository paper RH-354.
- [3] Bin Wang. A weighted prefix–tail gluing criterion for regularized determinants, 2026. Repository paper RH-288.
- [4] Bin Wang. A uniform linear-depth profile for the upper counterloop band, 2026. Repository paper RH-357.
- [5] Bin Wang. Logarithmic terminal-window accuracy thresholds for the complete upper counterloop band, 2026. Repository paper RH-359.
- [6] Bin Wang. A sharp post-first-alias mesoscopic crossover, 2026. Repository paper RH-356.
- [7] Bin Wang. Modulus-cap-forced growing-ladder signed cancellation, 2026. Repository paper RH-352.
- [8] Bin Wang. Ten layers of signed completion: Growing-depth coefficient-ledger underdetermination at the physical first alias, 2026. Repository paper RH-351.

- [9] Bin Wang. Synchronized determinant-prefix equivalence and a two-order orbit–head compensation obstruction, 2026. Repository paper RH-340.
- [10] Bin Wang. Terminal-lag geometric localization for the complete upper counterloop band, 2026. Repository paper RH-358.
- [11] Bin Wang. A terminal-lag exponential-tilt phase transition for the complete upper counterloop band, 2026. Repository paper RH-360.
- [12] Bin Wang. A ten-layer trace-envelope frontier review, 2026. Repository paper RH-241.
- [13] Bin Wang. Upper-alias counterloop burden and head-transfer precision, 2026. Repository paper RH-355.